

■ WATER CYCLE IN A BAG

A Complete Classroom Lesson Plan & Science Experiment Guide

■ Level

Grades 3–5 (Ages
8–11)

■ Duration

60–75 Minutes

■ Type

Hands-On
Experiment

■ Cost

Under ₹25

■ Origin

India-Contextual
Lesson

■ LESSON OVERVIEW

Lesson Description

In this exciting hands-on science lesson, students explore the water cycle by creating their own miniature model inside a sealed zip-lock bag. Through direct observation, students witness evaporation, condensation, and precipitation in action — all within a few hours. The lesson is deeply connected to India's monsoon season, making the science immediately relevant and relatable to students' everyday lives.

■ Learning Objectives

By end of lesson, students will be able to:

1. Name and explain the three stages of the water cycle: evaporation, condensation, and precipitation.
2. Demonstrate understanding of how the Sun drives the water cycle.
3. Construct a water cycle model using a zip-lock bag and observe changes over time.
4. Connect the water cycle to real-life Indian examples, including the monsoon.
5. Use scientific vocabulary accurately: water vapour, droplet, cycle, observe, heat.

Curriculum Connections

Subject	Topic / Standard	Relevance
Environmental Science	Water cycle, weather, natural resources	Core experiment topic
Geography	Monsoon, rivers (Ganga, Yamuna), Indian climate	Real-life application
Physics	States of matter, heat energy, phase changes	Evaporation & condensation
Language Arts	Scientific vocabulary, observation journaling	10 new science terms
Mathematics	Measuring 100ml water, time tracking	Measurement skills

■ BACKGROUND KNOWLEDGE FOR TEACHERS

The Water Cycle: A Complete Overview

The water cycle (also called the hydrological cycle) is the continuous movement of water within Earth and its atmosphere. It has no beginning or end — the same water molecules have been cycling through our planet for billions of years. The water you drink today may once have flowed through the Ganga river or fallen as snow on the Himalayas!

■■ EVAPORATION

Heat from the Sun gives energy to water molecules at the surface of oceans, lakes, rivers, and puddles. When molecules gain enough energy, they escape into the air as invisible water vapour (gas). This is why puddles disappear after a sunny day. In India, the intense summer heat causes massive evaporation from the Indian Ocean, fuelling the monsoon.

■ CONDENSATION

As water vapour rises into the cooler upper atmosphere, it loses heat energy and turns back into tiny liquid droplets. These droplets cluster around tiny dust particles to form clouds. You can see this process at home: a cold glass of lemon water on a humid day will have droplets forming on the outside — that is condensation in action.

■ PRECIPITATION

When water droplets in clouds combine and grow too heavy to stay aloft, they fall back to Earth as rain, hail, or snow. In India, the southwest monsoon (June–September) delivers about 70–80% of the country's annual rainfall — all powered by the same water cycle students will observe in their zip-lock bag.

The Role of the Sun

The Sun is the primary engine of the water cycle. It provides the heat energy necessary for evaporation. Without solar energy, water would remain as liquid in oceans and lakes with no mechanism to lift it into the atmosphere. India's location close to the equator means it receives intense solar radiation, which drives substantial evaporation from the warm Indian Ocean — this is the fundamental reason India experiences such dramatic seasonal monsoon rains.

India's Monsoon: The Water Cycle at Mega Scale

Stage	What Happens	Indian Example
Evaporation	Indian Ocean heats up in summer; massive water vapour rises	March–May heat over Bay of Bengal & Arabian Sea
Condensation	Vapour cools as it rises over Western Ghats and Himalayas	Bayam monsoon clouds form over Kerala from early J
Precipitation	Heavy rain falls across the subcontinent for 4 months	Kerala, Mumbai, Assam, Meghalaya (highest rainfa
Collection	Rain fills rivers, reservoirs, and groundwater	Ganga, Yamuna, Brahmaputra rivers swell; farmers

■ DETAILED LESSON PLAN (60–75 Minutes)

PHASE 1: HOOK & ENGAGEMENT

■ 5–8 min

1. Ask students: 'Yesterday there was a big puddle outside. Today it's gone. WHERE did the water go?'
2. Allow 2–3 students to share ideas. Accept all answers without correcting — this builds curiosity.
3. Introduce the water cycle chant: Say each word and do the action simultaneously.
4. EVAPORATION → raise both hands high above head
5. CONDENSATION → bring both hands slowly down in front of you
6. PRECIPITATION → wiggle fingers downward like falling rain
7. Practice the chant 3 times together as a class.

■ **Teacher**

Use a dramatic, story-teller voice when asking about the mystery puddle. The mystery hook engages students emotionally before the science begins.

Tip:

PHASE 2: DIRECT INSTRUCTION

■ 15 min

1. Write EVAPORATION, CONDENSATION, PRECIPITATION on the board.
2. EVAPORATION: Sun heats water → turns to invisible steam/vapour → floats up. Ask: 'Where do you see evaporation at home?' (Puddles drying, wet clothes drying, steam from chai/tea)
3. CONDENSATION: Water vapour cools → turns back to liquid droplets on a surface. Ask: 'Have you noticed your cold lemon water glass getting wet on the outside?'
4. PRECIPITATION: Clouds get heavy with droplets → rain (or snow) falls back down. Ask: 'Can you name the monsoon months in India?'
5. Draw a simple arrow cycle on the board: Sun → Evaporation (up) → Condensation (cloud) → Precipitation (down) → back to water body.
6. Explain: 'The Sun is the ENGINE that powers the whole cycle — without it, no rain!'

■ **Teacher**

Reference local examples specific to your region. Students in Tamil Nadu might relate to summer heat evaporating water faster. Use the Indian Ocean and monsoon connection to make the concept personal and memorable.

Tip:

PHASE 3: EXPERIMENT SETUP

■ 15 min

1. Distribute materials to each student or pair.
2. Step 1 — Decorate the bag: Using permanent marker, draw a bright sun at the TOP, fluffy clouds in the MIDDLE, wavy water lines at the BOTTOM.
3. Step 2 — Add water: Carefully pour exactly half a cup (100 ml) of water into the bag.
4. Step 3 — Add colour (optional): Put 1–2 drops of blue food colouring for visual effect.
5. Step 4 — Seal tightly: Zip the bag completely. Press out any extra air. Check for leaks!
6. Step 5 — Label the bag: Write student's name with permanent marker.
7. Step 6 — Mount the bag: Use cello tape to stick the bag firmly to a sunny window, sun drawing at top.
8. Tell students: 'Now we OBSERVE and WAIT. Record what you see in your notebook every 30 minutes.'

■ **Teacher**

Walk around to check all bags are properly sealed — a leaking bag is the most common problem. If there's no sunny window available, use a lamp placed very close to the bag. In Coimbatore / Tamil Nadu, windows face south for maximum sun exposure.

Tip:

PHASE 4: OBSERVATION & DISCUSSION

■ 15–20 min

1. After 30–45 minutes, have students observe their bags closely.
2. Ask guiding questions: 'Do you see tiny drops forming on the sides or top of the bag?'
3. Ask: 'Is the inside looking a little foggy? Where do you think that fog comes from?'
4. Ask: 'Are the drops running down the inside of the bag? What does that remind you of?'
5. Connect observations to vocabulary: 'The fog and drops forming — that's CONDENSATION! The drops running down — that's PRECIPITATION!'
6. Discuss: 'Notice the water at the bottom looks slightly less than before. The sun warmed it and some turned to vapour — that's EVAPORATION!'
7. Have students sketch the bag in their notebooks and label: evaporation arrows, cloud/condensation area, precipitation drops.

■ **Teacher**

If bags haven't shown much change after 30 minutes, move them to a brighter spot. The

Tip:

process is faster on sunny days. In Tamil Nadu's warm climate, results often appear within 20–25 minutes — faster than in cooler northern India!

PHASE 5: REFLECTION & WRAP-UP

■ 10 min

1. Ask 4–5 students to describe what they observed in their own words.
2. Review the key message: 'Water NEVER disappears — it just CHANGES FORM!'
3. Do the water cycle chant one final time.
4. Review all 10 vocabulary words: water cycle, evaporation, condensation, precipitation, water vapour, cloud, sun/heat, droplet, cycle, observe.
5. Assign take-home task: 'Leave your bag taped to a window overnight. Tomorrow morning, draw and describe what you see in your science notebook.'
6. Exit ticket (optional): 'On a small piece of paper, draw one stage of the water cycle and name it.'

■ **Teacher**

Encourage students to find their own examples of the water cycle at home over the next week and share in the next class. This reinforces learning through real-world observation.

■ MATERIALS & PREPARATION

Materials Per Student / Pair

Item	Quantity	Notes / Alternatives	Approx Cost
Clear zip-lock bag (medium/large)	1	Must be transparent; sandwich bag works well	■5–8
Water	100 ml (½ cup)	Plain tap water is fine	Free
Blue food colouring	1–2 drops	Optional but enhances visibility; can skip	■2
Permanent marker (black)	1 per student	Whiteboard marker also works on the outside	■10
Cello tape / masking tape	3–4 pieces	Enough to secure bag firmly to glass	■2
Sunny window or lamp	1 per class	South-facing window ideal; lamp 15–20 cm away	Free
Pencil and notebook	1 each	For observation recording and sketching	Free

Total cost per student: Under ■25 — making this one of the most affordable science experiments possible! All materials are readily available at any local stationery shop.

Teacher Preparation Checklist (Before Class)

- Collect all materials the day before and test that zip-lock bags don't leak.
- Identify the sunniest window in the classroom (south-facing is ideal in India).
- Pre-cut pieces of cello tape to save time during the experiment.
- If food colouring is available, prepare small cups or droppers for each group.
- Write the three vocabulary words on the board: EVAPORATION, CONDENSATION, PRECIPITATION.
- Optional: Prepare a simple diagram of the water cycle on chart paper to display during instruction.
- If doing with the whole class, consider doing a demo bag yourself the night before to have a 'finished' example ready.

■ KEY VOCABULARY

1. Water Cycle	The continuous movement of water from Earth's surface into the atmosphere and back again through evaporation, condensation, and precipitation.
2. Evaporation	The process by which liquid water is converted into water vapour (gas) when heated by the Sun.
3. Condensation	The process by which water vapour (gas) cools down and turns back into liquid water droplets, forming clouds or mist.
4. Precipitation	Water falling from clouds back to Earth's surface as rain, hail, sleet, or snow.
5. Water Vapour	Water in its gaseous (invisible) form, present in the air around us.

6. Cloud	A visible mass of tiny water droplets or ice crystals suspended in the atmosphere.
7. Sun / Heat	The Sun provides the energy (heat) that drives the water cycle by causing evaporation.
8. Droplet	A very small drop of water, such as those that form inside the zip-lock bag experiment.
9. Cycle	A series of events that repeats over and over in the same order, with no beginning or end.
10. Observe	To watch carefully and record what you notice using your senses during a scientific experiment.

■ ASSESSMENT & EVALUATION

Observation Journal Prompts

Students should record the following in their notebooks during and after the experiment:

Time 0 (Setup)	Describe your bag before placing it on the window. Draw and label it.
After 20–30 min	What do you notice? Are there any changes on the inside of the bag?
After 45–60 min	Describe the drops you see. Where are they forming? Are they moving?
Reflection 1	Which stage of the water cycle are you observing when drops form on the bag walls?
Reflection 2	How is your bag experiment similar to what happens in India during the monsoon?
Take-Home	Draw the bag the next morning. What has changed overnight? Explain using the three magic words.

Exit Ticket Questions (Formative Assessment)

Q1: Name the three stages of the water cycle in order.

Q2: What is the source of energy that makes the water cycle work?

Q3: When you see drops on a cold glass, which stage of the water cycle is that?

Q4: True or False: Water disappears from puddles forever.

Q5: Give one example of the water cycle that you see in your everyday life in India.

Rubric for Experiment Observation Record

Criteria	Excellent (4)	Good (3)	Developing (2)	Beginning (1)
Scientific vocabulary use	Uses all 3 terms correctly	Uses 2 terms correctly	Uses 1 term correctly	No terms used
Observation accuracy	Detailed, accurate, timestamped	Mostly accurate, some details	Basic observations only	Little observation recorded
Diagram quality	Clear, labelled, accurate	Mostly labelled	Some labelling	Diagram missing labels
India connection	Clearly explains monsoon link	Mentions India example	Vague India reference	No connection made

■ EXTENSION ACTIVITIES & DIFFERENTIATION

For Advanced Students

- Compare two bags: one placed in sun, one in shade. Record differences in condensation rate and write a hypothesis.
- Research the Cherrapunji / Mawsynram area in Meghalaya — the world's wettest place. Create a poster explaining WHY it rains so much using water cycle knowledge.
- Design a 'water cycle diagram' from scratch without looking at notes, and present it to the class.
- Research what 'transpiration' is and explain how it fits into the water cycle (evapotranspiration).

For Students Needing Support

- Provide a pre-drawn water cycle diagram with blanks to fill in the vocabulary words.
- Use visual flashcards: one picture per vocabulary word for reference during the activity.
- Pair with a stronger student for the experiment; encourage them to narrate observations aloud.

Cross-Curricular Extensions

LANGUAGE ARTS: Write a first-person story from the perspective of a water droplet travelling through the entire water cycle, starting as ocean water and ending as monsoon rain.

ART: Create a detailed water cycle mural for the classroom, including Indian landmarks (Indian Ocean, Western Ghats, Ganga river).

MATHEMATICS: Track rainfall data from two Indian cities (e.g., Mumbai and Jodhpur) over 12 months. Create a bar graph. Which city gets more rain? Why?

SOCIAL STUDIES: Discuss how farmers in India depend on monsoon rains for kharif crops. What happens when rains are late or inadequate?

Homework / Follow-up

- Leave the bag on the window overnight. Draw and describe the next morning.
- Find 3 examples of the water cycle at home and explain which stage each shows.
- Share the experiment with a family member and teach them the Water Cycle Chant.

■ ANTICIPATED STUDENT QUESTIONS & ANSWERS

Q: Why isn't my bag showing any drops yet?

A: The process needs warmth and time. Make sure the bag is in direct sunlight or close to a lamp. On cooler days or in air-conditioned rooms, it may take longer — up to 90 minutes. Be patient and keep observing!

Q: Why is the water staying blue at the bottom but the drops are clear?

A: Excellent observation! This is actually a bonus science lesson. When water evaporates, it leaves the food colouring behind. Only pure H₂O molecules turn to vapour — the dye molecules are too heavy to evaporate. The clear drops show that evaporation is a natural purification process!

Q: Is this the same as what happens to the ocean?

A: Exactly right! The ocean works the same way — just at a much, much larger scale. The Indian Ocean evaporates huge amounts of water every summer, which eventually becomes the monsoon rains. Your bag is a tiny model of this giant process.

Q: Will the water ever run out?

A: No! The water cycle keeps the same water moving in a loop forever. The water on Earth today is the same water that existed when dinosaurs lived. Water doesn't disappear — it just changes form and location.

Q: Why does it snow in Shimla but rain in Chennai?

A: Great question connecting geography! Temperature determines what form precipitation takes. In Shimla (high altitude, very cold), water freezes before reaching the ground, forming snow. In Chennai (low altitude, warm), it stays liquid and falls as rain.

■ SAFETY NOTES & CLASSROOM MANAGEMENT

Safety Notes

- Food colouring can stain clothing and surfaces — place newspaper under work area.
- Ensure bags are completely sealed before placing on windows to avoid water spills.
- If using a lamp instead of sunlight, keep it at a safe distance (15–20 cm) and ensure cables are away from walking paths.
- Permanent marker fumes — ensure classroom is ventilated.
- Supervise younger students when pouring water to avoid spills.

Classroom Management Tips

- Set up a 'science station' with materials pre-arranged before students arrive.
- Use a 'buddy system' — each pair has one 'materials manager' and one 'recorder'.
- Play soft background music during observation time to maintain a calm, focused atmosphere.
- Give a 2-minute warning before transitions to help students complete their observations.
- Create a class observation chart on the board where students can share findings.